

Eigenvalues and inverse problems

Conditioning of a random Krylov compression of a cyclic shift

Difficulty: challenging **Importance:** interesting to specialist**Status:** Open

Rating rationale: Challenging reflects a least-singular-value bound for a dependent random Krylov matrix; specialist impact concerns a structured compression mechanism in eigenvalue algorithms.

Problem statement

Let $C_n \in \mathbb{C}^{n \times n}$ be the cyclic shift, $C_n e_j = e_{j+1}$ for $j < n$ and $C_n e_n = e_1$. Draw b uniformly from the complex unit sphere. For $2 \leq k < n$, let Q have orthonormal columns spanning

$$\mathcal{K}_k(C_n, b) = \text{span}\{b, C_n b, \dots, C_n^{k-1} b\}, \quad H = Q^* C_n Q.$$

This subspace has dimension k almost surely. Define $\kappa_V(H) = \inf_{H=VDV^{-1}, D \text{ diagonal}} \|V\|_2 \|V^{-1}\|_2$, and set it to $+\infty$ if H is not diagonalizable. Do universal constants $C, c > 0$ exist such that, for every $n \geq 3$ and $2 \leq k < n$, with the same constants,

$$\Pr\{\kappa_V(H) \leq Cn^c\} \geq 0.99?$$

Changing the orthonormal basis of the same Krylov space does not affect this question. Randomness belongs to the starting vector; replacing Q by an independently Haar-distributed subspace would change the model.

Reference

Amsel et al., *Linear Systems and Eigenvalue Problems*, Problem 3.5, which explicitly singles out the circulant shift. This entry fixes “high probability” to a uniform 0.99 success target.

Earlier status check — 2026-09-08

Searches for *Krylov circulant shift condition 2026* and *random Krylov compression eigenvector condition number* found no solution. The deterministic starting vector e_1 gives a Jordan compression, so the unrandomized statement would be false; that exceptional example does not settle the probabilistic problem. The workshop report’s 20 August update reports [Peng’s obstruction](#) for arbitrary real diagonalizable inputs with ill-conditioned eigenvectors and real Gaussian starts. It explicitly leaves the normal-input case open, which includes the cyclic shift here.

Audit update — 2026-09-10

Rechecked [workshop version 3](#), Problem 3.5 and its August update. The arbitrary-diagonalizable counterexample does not cover this normal cyclic shift. Random Krylov/circulant conditioning searches found no bound resolving the displayed structured case.